

# Development of a Deep Reinforcement Learning Agent for Optimal Project Portfolio Budgeting under Uncertain Cash Flows

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## Abstract

Project portfolio budget allocation for EPC contractors in the oil and gas sector is a sequential decision-making problem under uncertain cash inflows, cash outflows, and project performance. Each portfolio consists of multiple projects with heterogeneous attributes, including budget at completion, start and finish dates, contractual S-curves, fixed profit margins, milestone-based payment structures, and uncertain contractor performance represented through Earned Value Management metrics such as cumulative SPI and CPI. The model also captures delayed and uncertain milestone payments, working-capital shortages, credit usage during negative-cashflow periods, and management intervention actions that affect cashflow and performance trajectories.

Despite its practical importance, rigorous mathematical modeling of project portfolio budgeting under joint cashflow and performance uncertainty remains largely absent in recent literature, while most AI/ML-related studies discuss the topic qualitatively. This paper proposes a custom stochastic reinforcement learning environment for learning adaptive budget allocation policies under unknown future uncertainty. A PPO-based agent is trained with a multi-objective reward function to allocate constrained budgets efficiently, prioritize economically viable projects, avoid loss-making allocations, and learn optimal allocation timing. The framework provides a foundational RL model for EPC oil and gas portfolio budgeting, supports future fine-tuning with anonymized industry data, and enables immediate next-period allocation recommendations as well as rolling 12-period trajectory simulation.

**Keywords:** Project Portfolio Budgeting, Reinforcement Learning, EPC Contractors, Oil and Gas Projects, Cashflow Uncertainty, Earned Value Management, PPO, Rolling Horizon Control

## 1 Introduction

### 1.0.1 Project definition

project is a style of getting things done. project is a temporary endeavor with a defined beginning and end. cite PMBOK from your neighborhood rapair shop to the power plant construction, there is an endeavour for creating product or service. there must be a mutually

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agreed upon scope for work. there must be a deadline defined for the project. we must plan the work ahead to get an estimate of the cost and the time required to complete the project. we must then allocate the resources to the project to get the project completed on time and within budget. we must then monitor the project to ensure that it is completed on time and within cost limits. we must then close the project and get the project completed on time and within cost limits.

now for an organization using this framework, they surely have multiple projects in their portfolio. they must then allocate the resources to the projects to get the projects completed on time and within cost limits. they must then monitor the projects to ensure that they are completed on time and within cost limits. they must then close the projects and get the projects completed on time and within cost limits.

but this approach at the high level board member annual budget allocation and strategic meeting has no established solution framework, not even a software to record the proper data.

there have been softwares introduced for the project management such as Microsoft Project, Primavera, and others. but these softwares are not designed to handle the complexity of the project portfolio management. they are designed to handle the complexity of the project management. they demonstrated features to capture the temporal progress of the projects and earned value metrics. they don't have the reasoning model to intelligently monitor projects. they just capture data. they are data entry forms. the first steps to digital maturity for management.

for the busy board member at headquarters managing different construction sites all over the contry and even abroad, these project control softwares just don't answer the need to give optimized budgeting solution.

there is some other functionality needed to be added on top of the management systems, to monitor the stream of data and get a complete view of the portfolio state and take budgeting allocation decisions.

even if somehow the company has the world's best portfolio manager he must have about 40 years of portfolio management, this is not comparable to the machine learning algorithm that could learn all the data for all the projects for decades of data. to be able to learn the data and generalize over the different states and environments, different macro and micro conditions.

we propose a configurable custome environment for the portfolio. user could configure the environment according to best fit to their experince and history, and use that environment to train an RL agent to propose the best allocations across the projects at each timestep.

we configure the environment according to our data providers and train RL agent on it. this RL agent's performance is compared to the conventional OR baseline solutions for proving efficiency. we stress test the environment to demonstrait the frameworks capability to include the most giving and the most punishing states and dynamics for the portfolio manager. the agent's superiority relative to the baselines through the stress testing prove the model's robustness.

## 1.1 Motivation and Context

Project portfolio management (PPM) is a fundamental organizational capability for maintaining competitive advantage and achieving strategic objectives. In project-based industries, portfolio decisions determine which initiatives are selected, prioritized, and funded under limited resources, conflicting criteria, and uncertain execution outcomes ([Archer and Ghasemzadeh, 1999](#); [Cooper et al., 2001](#); [Martinsuo, 2013](#); [Killen et al., 2008](#)). These

decisions become particularly critical in large-scale engineering environments, where organizations must coordinate multiple concurrent projects while balancing strategic value, financial feasibility, schedule pressure, and risk exposure.

In the Engineering, Procurement, and Construction (EPC) sector, especially in oil and gas projects, portfolio budgeting is complicated by heterogeneous contract structures, long project durations, capital intensity, and uncertain cash-flow behavior (Merrow, 2011; Cui et al., 2010; Kenley and Wilson, 1986; Kaka and Price, 1993). Budget allocation must account for both cash outflows associated with procurement, labor, equipment, and sub-contractors, and delayed cash inflows tied to contractual milestones and client approvals. Consequently, portfolio budgeting is not merely a one-time allocation problem; it is a sequential decision process in which current funding choices influence future project progress, liquidity conditions, and portfolio-level performance.

This challenge is amplified by persistent uncertainty in project execution. Large industrial and infrastructure projects frequently experience delays and cost overruns, and such deviations can propagate through both schedules and financial plans (Flyvbjerg et al., 2002, 2003; Love et al., 2013, 2016). Earned value indicators such as the Schedule Performance Index (SPI) and Cost Performance Index (CPI) provide a compact representation of execution performance and can support project monitoring and control (Fleming and Koppelman, 2016; Lipke et al., 2009; Vanhoucke, 2012). However, these indicators also underscore the difficulty of making reliable budget allocation decisions when future project states, payment realizations, and performance deviations remain uncertain. Therefore, portfolio budgeting under uncertain cash flows requires methods that can reason across multiple projects, multiple time periods, and stochastic state transitions.

The broader literature on project portfolio management similarly characterizes such environments by large data volumes, multiple decision criteria, resource constraints, and the need for investment optimization. These characteristics are consistent with prior research on project portfolio selection, project scheduling under uncertainty, and construction cash-flow forecasting (Gutjahr and Froeschl, 2013; Doerner et al., 2006; Khanzadi et al., 2018; Golenko-Ginzburg and Gonik, 1997; Ballestín and Blanco, 2015; Barraza et al., 2007). Taken together, they motivate the development of intelligent decision-support methods when static or manually driven budgeting approaches become insufficient.

## 1.2 Research Gap

A substantial body of literature has addressed project portfolio selection, prioritization, and scheduling through operations research, multi-criteria decision-making, stochastic programming, and metaheuristic optimization (Archer and Ghasemzadeh, 1999; Liesiö et al., 2007; Klerides and Hadjiconstantinou, 2019; Elshaer, 2014). These methods provide valuable structure for complex decision problems, but they are often less suitable when portfolio budget allocation must respond to evolving project conditions under uncertainty. In many existing formulations, project cash flows, budget requirements, and performance trajectories are treated as fixed inputs, deterministic plans, or exogenous scenarios. This limits their ability to capture the interaction between current allocation decisions and future project and financial states.

A central limitation in this context is that portfolio budgeting in EPC environments is inherently sequential. Budget releases made at one time step affect project execution, milestone realization, cash availability, and subsequent allocation opportunities. Yet much of the existing literature approaches project portfolio decisions as static optimization, stage-wise planning, or scheduling problems rather than as a learned sequential policy problem under uncertainty. As a result, the adaptive linkage between observed portfolio

state and subsequent allocation action remains insufficiently modeled.

Recent advances in machine learning and reinforcement learning provide a promising direction for this class of problems. Reinforcement learning is specifically designed for sequential decision-making under uncertainty and has demonstrated value in operations, scheduling, and process control settings (Puterman, 1994; Sutton and Barto, 2018; Wauters et al., 2016; Chen et al., 2018; Hubbs et al., 2020; Bengio et al., 2021). Nevertheless, its application to project portfolio budgeting under uncertain cash flows remains limited. There is still a need for a reproducible framework that can integrate project-level uncertainty, cash-flow timing, milestone-based payments, and adaptive resource allocation within a portfolio-level sequential decision architecture.

A further practical constraint is data availability. Detailed industrial project portfolio data are typically confidential and difficult to release publicly, which creates a barrier to reproducible experimentation. As a result, the literature lacks sufficiently transparent benchmark environments for testing adaptive portfolio budgeting policies. This gap motivates the use of literature-calibrated synthetic data as a practical foundation for developing and evaluating learning-based decision-support systems (Jordon et al., 2022; Xu et al., 2019).

### 1.3 Proposed Approach and Contributions

To address these gaps, this paper develops a deep reinforcement learning framework for optimal project portfolio budgeting under uncertain cash flows. The problem is modeled as a sequential decision process in which an agent observes the evolving state of a project portfolio and allocates constrained financial resources over time. The state representation incorporates project-level and portfolio-level information, including budget at completion, planned duration, contractual cash-flow patterns, performance indicators, milestone-based payment structures, and liquidity-related variables.

The framework is designed to learn an allocation policy directly from repeated interaction with a stochastic simulation environment. Rather than relying on predefined deterministic decision rules or repeated re-optimization procedures, the agent learns how present allocation choices affect future portfolio states and cumulative performance over time. To learn this policy, the study employs deep reinforcement learning, with Proximal Policy Optimization (PPO) as the primary training algorithm (Schulman et al., 2017), consistent with the broader reinforcement-learning literature on stochastic control and policy optimization (Sutton and Barto, 2018; Haarnoja et al., 2018).

After training, the learned policy can also be deployed within the simulation environment to generate a forward policy rollout over a specified horizon, such as the next 12 time steps, in order to illustrate plausible portfolio evolution and corresponding recommended allocations. This forward policy rollout should be interpreted as a simulated policy-driven trajectory rather than as a realized future observation. Its purpose is to demonstrate the behavioral implications of the learned allocation policy under the modeled environment dynamics until actual future project states become available.

Because real EPC portfolio datasets are typically inaccessible due to confidentiality restrictions, the framework uses a literature-calibrated synthetic data generation process. This allows reproducible experimentation while preserving consistency with known characteristics of project cash flows, schedule variability, and portfolio structure. The synthetic environment is intended as a research testbed that can later be refined and calibrated using anonymized organizational data.

The main contributions of this paper are as follows:

1. **A sequential formulation of project portfolio budgeting:** The study formulates EPC portfolio budgeting under uncertain cash flows as a dynamic decision problem in which allocations, project progress, and liquidity conditions evolve jointly over time.
2. **A custom reinforcement learning environment for portfolio budgeting:** The paper develops an environment that captures project-level uncertainty, cash inflows and outflows, milestone-based payments, and portfolio-level financial constraints.
3. **A deep reinforcement learning policy for adaptive allocation:** The framework trains an agent to learn budget allocation decisions directly from stochastic interaction with the environment.
4. **A forward policy rollout capability for trajectory demonstration:** The study shows how the trained policy can be used to simulate plausible future portfolio trajectories and recommended allocations over a finite horizon.
5. **A literature-calibrated synthetic data framework:** The study provides a reproducible synthetic portfolio-generation procedure to address the industrial data confidentiality barrier.

## 1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews prior research on project portfolio management, project control under uncertainty, and reinforcement learning for sequential decision-making. Section 3 formulates the project portfolio budgeting problem under uncertain cash flows as a sequential stochastic decision problem. Section 5 presents the proposed solution methodology, including the synthetic data generation framework, cash-flow and uncertainty modeling, reinforcement learning environment design, and PPO-based training procedure. Section 6 reports the computational results and analysis, including benchmark comparisons, robustness assessment, and forward policy rollout analysis. Finally, Section 7 concludes the paper and outlines future research directions.

# 2 Literature Review

## 2.1 Project Management

### 2.1.1 Cashflow Modeling and S-Curves

Project cashflow modeling has a rich history in construction management. [Kenley and Wilson \(1986\)](#) introduced the idiographic approach to cashflow forecasting, recognizing that each project exhibits unique spending patterns. [Kaka and Price \(1993\)](#) developed neural network models for cashflow prediction based on historical project data.

The S-curve model, representing cumulative spending as a sigmoid function of project duration, has become the standard representation. [Cioffi \(2005\)](#) provided analytical parameterization of S-curves using Beta distributions, enabling closed-form expressions for cashflow velocity and peak spending timing. [Barraza et al. \(2007\)](#) extended this to probabilistic S-curves incorporating uncertainty in project duration and spending patterns.

[Cui et al. \(2010\)](#) analyzed milestone-based payment structures in construction contracts, finding that 87% of contracts use progress-based payments rather than time-based

schedules. This finding motivates our modeling choice to link payment timing to actual progress (SPI) rather than calendar dates.

### 2.1.2 Advance Payment and Recovery in EPC Contracts

Advance payment is a common contractual and financial mechanism in EPC and EPC/Turnkey projects, particularly in capital-intensive sectors such as oil and gas, where contractors often face substantial early cash requirements for mobilization, engineering start-up, procurement, equipment ordering, and site preparation. Standard EPC contract forms, including the FIDIC Silver Book, recognize advance payment as an early disbursement made by the Employer to the Contractor, normally subject to the provision of an Advance Payment Guarantee [FIDIC \(1999, 2017\)](#). This guarantee protects the Employer because the payment is made before the corresponding portion of the works has been fully executed or certified. From a contract-management perspective, the advance payment should not be interpreted as an additional payment beyond the Contract Price; rather, it represents an early release of part of the Contract Price to support the contractor’s working-capital needs [QuollNet \(nd\)](#); [International Construction Knowledge Hub \(nd\)](#). Accordingly, the advance is typically recovered through deductions from subsequent interim or milestone payment certificates, with the recovery schedule defined in the contract conditions, Particular Conditions, or Contract Data [FIDIC \(1999, 2017\)](#). This design allows the contractor to benefit from early liquidity while ensuring that the Employer does not pay more than the agreed Contract Price over the project lifecycle. Therefore, in EPC project financial management, advance payment and its recovery primarily affect the timing and distribution of cash flows rather than the total contractual revenue. This distinction is particularly important in portfolio-level project control, where early liquidity support, delayed net receipts during recovery periods, and the interaction with external financing can materially influence the contractor’s short-term financial balance and budget allocation decisions.

### 2.1.3 Project Performance Uncertainty and Empirical Evidence

A substantial body of empirical research documents systematic cost and schedule overruns in capital projects. [Flyvbjerg et al. \(2002\)](#) analyzed 258 transportation infrastructure projects, finding mean cost overruns of 28% and schedule delays of 27%. [Flyvbjerg et al. \(2003\)](#) extended this analysis to 258 projects across multiple sectors, confirming left-skewed performance distributions with long tails toward poor outcomes.

[Merrow \(2011\)](#) provided the most comprehensive analysis of EPC project performance, examining 318 industrial megaprojects in oil & gas, chemicals, and mining. Key findings include: (1) mean SPI at completion of 0.81 (19% behind schedule), (2) mean CPI of 0.87 (15% over budget), (3) positive correlation between schedule and cost performance ( $\rho \approx 0.68$ ), and (4) systematic variation in performance by project size, complexity, and contractor experience.

[Love et al. \(2013\)](#) fitted probability distributions to schedule overrun data from 276 construction projects, validating Beta and lognormal distributions as appropriate models. [Love et al. \(2016\)](#) analyzed the relationship between SPI and CPI, confirming that schedule delays typically precede cost overruns due to extended overhead and acceleration costs.

## 2.2 Portfolio Management

### 2.2.1 Project Portfolio Management Under Uncertainty

Project portfolio management (PPM) has been extensively studied in operations research and management science. Early work by [Archer and Ghasemzadeh \(1999\)](#) established frameworks for project selection and prioritization, while [Cooper et al. \(2001\)](#) developed stage-gate processes for new product development portfolios. However, these approaches primarily address the *selection* problem (which projects to pursue) rather than the *execution* problem (how to allocate resources dynamically).

[Gutjahr et al. \(2008\)](#) formulated multi-objective portfolio optimization models balancing expected return, risk, and strategic alignment. [Doerner et al. \(2006\)](#) applied Pareto ant colony optimization to generate efficient frontiers for portfolio selection. These methods assume deterministic project execution once selected, ignoring the sequential decision-making required under performance uncertainty.

Stochastic programming approaches have been proposed to handle uncertainty. [Golenko-Ginzburg and Gonik \(1997\)](#) modeled resource-constrained project scheduling with stochastic activity durations. [Klerides and Hadjiconstantinou \(2019\)](#) developed two-stage stochastic programs for portfolio selection under demand uncertainty. However, these methods require explicit scenario trees and known probability distributions, which are unavailable for contractor performance dynamics.

## 2.3 Reinforcement Learning in Operations Research

RL has emerged as a powerful tool for sequential decision-making under uncertainty. [Sutton and Barto \(2018\)](#) provide the foundational textbook covering Markov Decision Processes (MDPs), temporal difference learning, and policy gradient methods. Recent advances in deep RL, particularly Proximal Policy Optimization (PPO) ([Schulman et al., 2017](#)) and Soft Actor-Critic (SAC) ([Haarnoja et al., 2018](#)), have enabled application to high-dimensional continuous control problems.

Applications of RL to operations research problems have grown rapidly. [Hubbs et al. \(2020\)](#) applied deep RL to chemical production scheduling, achieving near-optimal solutions with 1000x speedup over MIP. [Bengio et al. \(2021\)](#) surveyed machine learning for combinatorial optimization, highlighting RL’s ability to learn heuristics that generalize across problem instances.

In project management, RL applications remain limited. [Wauters et al. \(2016\)](#) used RL for resource-constrained project scheduling with deterministic durations. [Chen et al. \(2018\)](#) applied RL to construction project scheduling under resource uncertainty. However, no prior work addresses portfolio-level budgeting under cashflow uncertainty with stochastic contractor performance.

## 2.4 Synthetic Data Generation for OR/ML Research

The use of synthetic data in operations research has a long tradition. [Kolisch and Sprecher \(1997\)](#) created the Project Scheduling Problem Library (PSPLIB), a benchmark dataset of synthetic instances that became the standard for algorithm comparison. [?](#) generated synthetic project networks to study schedule risk analysis methods.

In machine learning, synthetic data generation has become essential for domains with data scarcity or privacy constraints. [Jordon et al. \(2022\)](#) surveyed synthetic data generation methods, emphasizing the importance of preserving statistical properties of real data.

Xu et al. (2019) developed generative models for tabular data that maintain correlations and distributional characteristics.

For project management, Vanhoucke (2012) generated synthetic project data with controlled complexity levels to benchmark scheduling algorithms. Elshaer (2014) created synthetic portfolios with varying risk profiles to test portfolio optimization methods. Our work extends this tradition by developing a comprehensive synthetic data generator calibrated to empirical distributions from multiple literature sources, ensuring realism while maintaining reproducibility.

## 2.5 Research Gaps

Despite extensive research in project portfolio management, performance uncertainty, and RL applications, critical gaps remain:

1. **No RL framework for portfolio budgeting:** Existing RL applications to project management focus on single-project scheduling, not portfolio-level resource allocation under cashflow uncertainty.
2. **Lack of realistic synthetic environments:** Prior synthetic data generators for project management do not model stochastic contractor performance, milestone-based payments, or portfolio-level cashflow dynamics.
3. **Absence of rolling horizon RL architectures:** Rolling horizon control has not been integrated with RL for portfolio management, despite its natural fit for handling variable-duration portfolios.
4. **No methodology for literature-calibrated data generation:** Existing approaches either use proprietary data (unavailable for research) or arbitrary synthetic distributions (lacking empirical grounding).

This paper addresses these gaps by developing the first RL framework for portfolio budgeting with a literature-calibrated synthetic environment and rolling horizon architecture.

## 3 Problem Formulation

## 4 Problem Formulation

### 4.1 Overview

This section develops the formal problem structure for portfolio-level budget allocation in contracted project portfolios. The formulation proceeds in three levels of increasing generality, each building directly on the preceding one.

**Level 1 — Deterministic Full-Foresight Linear Program.** All contractor performance parameters, expenditure trajectories, and payment timing are treated as known with certainty. The resulting program is a pure linear program (LP) in the continuous budget allocation variables. Because no real decision-maker possesses this information ex ante, the Level 1 optimum constitutes a theoretical upper bound on portfolio performance: no implementable policy can exceed it. This upper bound anchors the empirical comparison in Section ??.

**Level 2 — Stochastic Extension and Intractability.** Deterministic performance parameters are replaced by stochastic processes consistent with the empirical EVM literature. The resulting multi-stage stochastic program is formally characterized, and the combinatorial growth of its scenario tree with the number of projects and planning periods is established. This intractability argument motivates the reformulation in Level 3.

**Level 3 — Constrained Markov Decision Process.** The stochastic program of Level 2 is reformulated as a Constrained Markov Decision Process (CMDP). An explicit mapping is provided from LP elements to MDP primitives: decision variables map to actions, constraint parameters map to state variables, the objective maps to the reward function, and feasibility constraints define the constrained action space. The CMDP is implemented as a configurable, Gymnasium-compatible synthetic environment; a Proximal Policy Optimization (PPO) agent trained on this environment serves as the primary computational demonstration of the framework’s utility.

The three levels are not alternative models of the same problem. They are a coherent modeling arc: Level 1 defines the problem’s theoretical ceiling, Level 2 characterizes why that ceiling is computationally unreachable, and Level 3 provides the tractable reformulation that makes the problem solvable at realistic scale. The remainder of this section develops each level in turn, preceded by the shared problem primitives — project and portfolio definitions, contractual cash-flow mechanics, and S-curve expenditure dynamics — on which all three levels draw.

The formulation integrates domain knowledge and established methodologies from project management and operations research: S-curve expenditure dynamics [Barraza et al. \(2004\)](#), Earned Value Management (EVM) performance tracking [Fleming and Koppelman \(2016\)](#), milestone-based payment structures consistent with FIDIC contract forms [FIDIC \(2017\)](#); [?](#); [?](#), and the theory of investment under uncertainty as it applies to abandonment options [?](#). The LP of Level 1 is situated within the stochastic programming tradition [Birge and Louveaux \(2011\)](#); the CMDP of Level 3 is situated within the constrained MDP literature [?](#), framed as a direct descendant of Bellman’s dynamic programming and stochastic control theory rather than as an import from machine learning.

## 4.2 Scope and Positioning

The problem addressed here is operational, not strategic. The literature on project portfolio management has devoted considerable attention to the *selection* problem: given a candidate set of projects and a capital budget, which subset should the organization commit to [???](#) This work takes the selection decision as given. A portfolio of contracted projects exists; commitments have been signed; and the relevant decision-maker — a portfolio manager or governing board — faces a recurring operational question: how should the available budget be allocated across active projects at each discrete planning period?

This is the budget allocation control problem, and it is the sole focus of the formulation. The decision-maker exercises one lever: the amount of budget directed to each active project at each period. All strategic and operational outcomes of interest — milestone achievement, cash flow balance, schedule adherence, and ultimate project success or termination — flow through this single allocation decision. The formulation makes this causal chain explicit and embeds it within a sequential decision-making framework.

The model is deliberately general. Every structural element — the number of projects, the milestone structure, the S-curve shape parameters, the advance payment and retention ratios, the performance response distribution, the deviation penalty weight, and the termination settlement mechanics — is parameterized rather than fixed. Sector-specific instantiation is achieved through empirical calibration of these parameters, not through

changes to the model structure. The calibration used in this work draws on anonymized data from two major EPC contractors, supplemented by World Bank ICR data ? and infrastructure cost overrun datasets Flyvbjerg et al. (2003). The general structure is applicable across construction, software development, defense contracting, and other project-intensive sectors ??.

One deliberate boundary requires explicit statement. The model represents the financial relationship between a single client organization (the portfolio owner) and its contracted project delivery partners (the contractors). The client allocates budget; the contractor expends it and delivers progress. No third-party management consultant, program management contractor, or intermediary role is modeled. This two-party structure is the minimal representation consistent with the contractual mechanics encoded in the formulation and is retained across all three levels. Extensions to multi-party governance structures are noted as a direction for future work.

### 4.3 Fundamental Modeling Choices

Four structural choices underlie the entire formulation. Each is stated here with its justification, because they determine the form of every constraint and variable that follows.

**A portfolio is an aggregation of independent bilateral contracts.** Each project in the portfolio corresponds to a separate contract between the client and a distinct contractor. Projects do not share resources, workforces, or contractual obligations within the formulation; interaction occurs only through the shared portfolio budget constraint. This independence structure is consistent with standard EPC contracting practice and permits the portfolio problem to be decomposed into project-level dynamics aggregated at the portfolio level. It is also the structure that makes the LP of Level 1 tractable: the coupling across projects enters only through the single budget constraint at each period.

**The resource allocation problem reduces to budget allocation.** Within each project, the contractor controls workforce, materials, equipment, and subcontracting. From the portfolio manager’s perspective, the only instrument available is the budget directed to each project. The contractor’s internal response to that budget — the efficiency with which it is converted into physical progress — is modeled as a performance parameter, stochastic at Level 2 and treated as known at Level 1. This reduction of the resource allocation problem to a scalar budget allocation per project per period is both a modeling choice and a reflection of the actual decision authority of a portfolio-level governing body.

**Time is discrete, with periods representing calendar months.** The planning horizon  $\mathcal{T} = \{1, \dots, H\}$  is indexed by discrete periods. Monthly periods are adopted because the key governance events in portfolio management — progress reporting, payment certification, performance review meetings — are conventionally held on a monthly cycle. The LP of Level 1 optimizes over the full horizon simultaneously; the CMDP of Level 3 operates period by period, with decisions made sequentially as new information is revealed.

**Contract mechanics are calibrated to FIDIC forms.** The milestone-based payment structure, advance payment and recovery mechanism, retention and release, and termination settlement mechanics are all specified to be consistent with FIDIC Red, Yellow, and Silver Book contract forms FIDIC (2017). This anchors the formulation in the dominant

international EPC contracting framework and ensures that the model parameters have direct empirical counterparts in contractor financial data. Where FIDIC allows variation by agreement (e.g., the advance payment ratio, the retention percentage, the deviation penalty weight), these are treated as project-specific calibrated parameters rather than fixed constants.

#### 4.4 Problem Primitives

This subsection establishes the shared notation, project-level parameters, contractual cash-flow mechanics, and portfolio aggregation structure on which all three levels of the formulation draw. Nothing introduced here is level-specific; each symbol defined below appears, without redefinition, in the LP of Level 1, the stochastic extension of Level 2, and the CMDP of Level 3.

##### 4.4.1 Sets and Indices

$\mathcal{P} = \{1, \dots, n\}$  Set of projects in the portfolio, indexed by  $i$ . All projects are assumed committed at the start of the planning horizon; no new projects enter mid-horizon.

$\mathcal{T} = \{1, \dots, H\}$  Set of discrete planning periods, indexed by  $t$ , each representing one calendar month.

$\mathcal{M}_i = \{1, \dots, M_i\}$  Ordered set of contractual milestones for project  $i$ , indexed by  $j$ . Milestones are ordered by their progress thresholds:  $\theta_{i,1} < \theta_{i,2} < \dots < \theta_{i,M_i} = 1$ .

$\mathcal{A}_i^{\text{plan}} = \{t : s_i \leq t \leq f_i\}$  Planned execution window of project  $i$ , determined by the planned start  $s_i$  and planned finish  $f_i$ .

$\mathcal{P}_t^{\text{active}} \subseteq \mathcal{P}$  Subset of projects active at period  $t$ . In Level 1 (full foresight), this set is determined ex ante from the known efficiency trajectories  $\{\bar{\eta}_{i,t}\}$  and is treated as a fixed parameter. In Level 2 and Level 3, it is endogenous, depending on realized project status transitions.

$\mathcal{M}_i^{\text{cert}}(t) \subseteq \mathcal{M}_i$  Subset of milestones of project  $i$  certified up to and including period  $t$ :  
 $\mathcal{M}_i^{\text{cert}}(t) = \{j \in \mathcal{M}_i : \theta_{i,j} \leq P_{i,t}^{\text{actual}}\}.$

##### 4.4.2 Project-Level Parameters

Each project  $i \in \mathcal{P}$  is fully characterized at commitment by the contractual and operational parameters in Table 1. These parameters are fixed at the start of the planning horizon and do not change during execution.

| Parameter                           | Symbol              | Domain           | Definition                                                                                                                                                                                             |
|-------------------------------------|---------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Planned start &amp; finish</b>   | $s_i, f_i$          | $\mathbb{Z}^+$   | Planned start and finish periods for project $i$ .                                                                                                                                                     |
| <b>Planned duration</b>             | $D_i^{\text{plan}}$ | $\mathbb{Z}^+$   | $D_i^{\text{plan}} = f_i - s_i + 1$ .                                                                                                                                                                  |
| <b>Budget at completion</b>         | $\text{BAC}_i$      | $\mathbb{R}^+$   | Total planned expenditure (contractor cost basis).                                                                                                                                                     |
| <b>Contract price</b>               | $CP_i$              | $\mathbb{R}^+$   | Total payment from client: $CP_i = \text{BAC}_i(1 + \pi_i)$ .                                                                                                                                          |
| <b>Profit margin</b>                | $\pi_i$             | $\mathbb{R}_0^+$ | Contractual markup on $\text{BAC}_i$ .                                                                                                                                                                 |
| <b>Advance payment ratio</b>        | $\alpha_i$          | $[0, 1]$         | Fraction of $CP_i$ disbursed at project initiation; $A_i = \alpha_i CP_i$ .                                                                                                                            |
| <b>Retention ratio</b>              | $\rho_i$            | $[0, 1]$         | Fraction withheld from each milestone payment; released upon project completion.                                                                                                                       |
| <b>Milestone threshold</b>          | $\theta_{i,j}$      | $(0, 1]$         | Cumulative progress fraction required to certify milestone $j$ .                                                                                                                                       |
| <b>Milestone payment weight</b>     | $\phi_{i,j}$        | $\mathbb{R}^+$   | Fraction of $CP_i$ released at milestone $j$ ; $\sum_{j \in \mathcal{M}_i} \phi_{i,j} = 1$ .                                                                                                           |
| <b>Deviation penalty weight</b>     | $\lambda_i$         | $\mathbb{R}_0^+$ | Per-period cost per unit of signed progress deviation from plan; symmetric, capped, and applied continuously. Calibrated to reflect the contractual delay penalty rate $d_i$ at the schedule boundary. |
| <b>Deviation penalty cap</b>        | $\Lambda_i$         | $\mathbb{R}_0^+$ | Maximum cumulative deviation penalty as a fraction of $CP_i$ ; mirrors the contractual delay penalty cap and applies symmetrically to schedule underrun and overrun.                                   |
| <b>S-curve shape parameters</b>     | $a_i, b_i$          | $\mathbb{R}^+$   | Shape parameters of the regularized incomplete Beta function $F_i(\tau) = I_\tau(a_i, b_i)$ governing the planned expenditure profile.                                                                 |
| <b>Termination threshold</b>        | $\kappa_i$          | $\mathbb{R}_0^+$ | Cumulative deviation penalty level at which the project becomes a candidate for termination; defined as a fraction of $\Lambda_i$ .                                                                    |
| <b>Mandatory minimum allocation</b> | $m_{i,t}^0$         | $\mathbb{R}_0^+$ | Minimum budget allocation required to maintain contractual obligations for project $i$ at period $t$ .                                                                                                 |
| <b>Allowable delay buffer</b>       | $\text{buffer}_i$   | $\mathbb{Z}_0^+$ | Number of periods beyond $f_i$ during which project $i$ remains active before forced termination is assessed.                                                                                          |
| <b>Discount factor</b>              | $\gamma$            | $(0, 1]$         | Per-period discount factor applied to future cash flows.                                                                                                                                               |
| <b>Initial cash balance</b>         | $B_1$               | $\mathbb{R}^+$   | Portfolio cash balance at the start of the planning horizon.                                                                                                                                           |

Table 1: Project-level and portfolio-level parameters. All parameters are fixed at commitment and calibrated empirically; none are decision variables.

#### 4.4.3 Planned Expenditure Profile (S-Curve)

The planned cumulative expenditure of project  $i$  follows a normalized S-curve ?Barraza et al. (2004). Define the normalized planned time at period  $t$  as:

$$\tau_{i,t} = \frac{t - s_i + 1}{D_i^{\text{plan}}}, \quad t \in \mathcal{A}_i^{\text{plan}},$$

with truncation  $\tau_{i,t}^c = \min\{1, \max\{0, \tau_{i,t}\}\}$  to handle periods outside the planned window. The planned cumulative expenditure fraction is:

$$F_i(\tau) = I_\tau(a_i, b_i),$$

where  $I_\tau(\cdot, \cdot)$  is the regularized incomplete Beta function and  $a_i, b_i > 0$  are empirically calibrated shape parameters. The planned cumulative expenditure and planned period expenditure are:

$$\bar{E}_{i,t} = \text{BAC}_i \cdot F_i(\tau_{i,t}^c), \quad \Delta \bar{E}_{i,t} = \bar{E}_{i,t} - \bar{E}_{i,t-1}.$$

The planned cumulative progress tracks planned expenditure:

$$P_{i,t}^{\text{plan}} = F_i(\tau_{i,t}^c).$$

#### 4.4.4 State Variables

At each period  $t$ , the state of project  $i$  is characterized by the following variables. In Level 1 these are deterministic functions of the allocation decisions  $\{x_{i,t}\}$  under known efficiency parameters. In Level 2 and Level 3 they are stochastic.

$P_{i,t}^{\text{actual}} \in [0, 1]$  Cumulative actual progress of project  $i$  at the end of period  $t$ : the fraction of project scope physically completed.

$E_{i,t} \geq 0$  Cumulative actual expenditure on project  $i$  up to and including period  $t$ .

$\text{SPI}_{i,t}$  Schedule Performance Index at period  $t$ :

$$\text{SPI}_{i,t} = \frac{P_{i,t}^{\text{actual}}}{P_{i,t}^{\text{plan}}}, \quad P_{i,t}^{\text{plan}} > 0.$$

Values below unity indicate schedule underperformance; values above unity indicate acceleration ahead of plan.

$\text{CPI}_{i,t}$  Cost Performance Index at period  $t$ :

$$\text{CPI}_{i,t} = \frac{P_{i,t}^{\text{actual}} \cdot \text{BAC}_i}{E_{i,t}}, \quad E_{i,t} > 0.$$

Values below unity indicate cost overrun relative to work performed.

$\delta_{i,t}$  Signed progress deviation at period  $t$ :

$$\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}}.$$

Negative values indicate schedule underrun; positive values indicate schedule overrun ahead of plan.

$\Delta_{i,t}^{\text{pen}} \geq 0$  Cumulative deviation penalty accrued by project  $i$  up to period  $t$ , defined in Section 4.4.7.

$B_t \geq 0$  Available portfolio cash balance at the beginning of period  $t$ .

$\text{status}_{i,t} \in \{\text{active, completed, terminated}\}$  Categorical project status at period  $t$ . Transitions are governed by the progress recursion and the termination conditions defined in Section 4.4.8.

#### 4.4.5 Budget-to-Progress Conversion

The decision variable  $x_{i,t} \geq 0$  is the budget allocated to project  $i$  at period  $t$ . Allocated budget is converted to a progress increment through an efficiency parameter:

$$\Delta P_{i,t}^{\text{actual}} = \eta_{i,t} \cdot \frac{x_{i,t}}{\text{BAC}_i},$$

where  $\eta_{i,t}$  is the *budget-to-progress efficiency parameter* for project  $i$  at period  $t$ . This parameter captures all sources of conversion uncertainty — workforce productivity, supply chain conditions, technical difficulty, weather — that determine how much physical progress a given budget expenditure yields.

**Treatment across levels.** In Level 1,  $\eta_{i,t}$  is replaced by a known deterministic value  $\bar{\eta}_{i,t}$ , calibrated from the project’s planned performance profile. Under the nominal plan,  $\bar{\eta}_{i,t} = 1$  implies that budget expenditure tracks the S-curve exactly; deviations encode a known performance trajectory. In Level 2 and Level 3,  $\eta_{i,t}$  is a random primitive realized only after  $x_{i,t}$  is committed, with a distribution calibrated from empirical EVM data.

The cumulative state recursion is:

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \eta_{i,t} \cdot \frac{x_{i,t}}{\text{BAC}_i}, \quad E_{i,t} = E_{i,t-1} + x_{i,t},$$

with  $P_{i,0}^{\text{actual}} = 0$  and  $E_{i,0} = 0$ .

#### 4.4.6 Contractual Cash-Flow Mechanics

Project cash flows consist of outflows (budget allocations  $x_{i,t}$ ) and inflows (contractual payments). The payment structure is consistent with FIDIC contract forms [FIDIC \(2017\)](#) and comprises three mechanisms: advance payment with progressive recovery, interim milestone-based payments with retention, and final payment with retention release.

**Advance Payment.** At project initiation (period  $s_i$ ), the client disburses an advance payment:

$$A_i = \alpha_i \cdot CP_i.$$

The advance payment is a contractual timing adjustment, not a financing instrument: it carries no interest obligation and is not recorded as debt ???. It is recovered progressively through deductions from subsequent milestone payments.

**Advance Recovery.** The recovery deduction applied at milestone  $j$  is proportional to the milestone payment weight:

$$\Delta A_{i,j} = \phi_{i,j} \cdot A_i.$$

This ensures the full advance is recovered by the time all milestones are certified, regardless of project duration.

**Milestone Certification and Net Payment.** Milestone  $j \in \mathcal{M}_i$  is certified at the first period  $t$  such that  $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$ . On certification:

$$R_{i,j} = \phi_{i,j} \cdot CP_i \cdot (1 - \rho_i), \quad R_{i,j}^{\text{net}} = R_{i,j} - \Delta A_{i,j}.$$

Advance payment recovery and retention operate independently: recovery deductions are computed on gross payments prior to retention, and retained amounts accumulate separately until released at completion. The full accounting identity across all payments is:

$$A_i + \sum_{j \in \mathcal{M}_i} R_{i,j}^{\text{net}} + R_i^{\text{ret}} = CP_i,$$

where  $R_i^{\text{ret}} = \rho_i \cdot CP_i$  is the total retention released at completion.

**Final Payment and Retention Release.** Upon project completion ( $P_{i,t}^{\text{actual}} = 1$ ), the retained amount is released. The final net payment is:

$$R_i^{\text{final}} = R_{i,M_i}^{\text{net}} + R_i^{\text{ret}}.$$

No separate delay penalty term appears here; schedule adherence is enforced entirely through the continuous running deviation penalty defined in Section 4.4.7. The retention release and final milestone payment together constitute the natural completion incentive embedded in the payment structure.

**Termination Settlement.** If project  $i$  is terminated at period  $t_i^{\text{term}}$  before completion, the net settlement payment is:

$$V_i^{\text{term}} = P_{i,t_i^{\text{term}}}^{\text{actual}} \cdot CP_i, \quad A_i^{\text{outstanding}} = A_i - \sum_{j \in \mathcal{M}_i^{\text{cert}}(t_i^{\text{term}})} \Delta A_{i,j}, \quad R_i^{\text{term}} = V_i^{\text{term}}(1 - \rho_i) - A_i^{\text{outstanding}}.$$

If  $R_i^{\text{term}} < 0$ , the outstanding advance payment balance constitutes a net obligation settled from available portfolio cash inflows ahead of other commitments, consistent with FIDIC termination clauses [FIDIC \(2017\)](#).

#### 4.4.7 Symmetric Running Deviation Penalty

Schedule adherence is enforced through a continuous symmetric deviation penalty applied at every period  $t \in \mathcal{T}$ . The per-period penalty for project  $i$  is:

$$c_{i,t}^{\text{dev}} = \min\{\lambda_i \cdot |\delta_{i,t}|, \Lambda_i \cdot CP_i\},$$

where  $\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}}$  is the signed progress deviation,  $\lambda_i \geq 0$  is the per-unit deviation penalty weight, and  $\Lambda_i$  is the penalty cap expressed as a fraction of  $CP_i$ .

Three design properties of this formulation warrant explicit statement.

**Symmetry.** The penalty is applied to  $|\delta_{i,t}|$ , not to  $\max\{0, -\delta_{i,t}\}$ . Both schedule underrun ( $\delta_{i,t} < 0$ ) and schedule overrun ( $\delta_{i,t} > 0$ ) are penalized equally. This discourages the decision-maker from accelerating projects beyond the planned profile, which would consume budget inefficiently and risk scope or quality degradation [Merrow \(2011\)](#).

**Continuity.** The penalty is assessed at every period, not only at contractual completion. This connects schedule adherence directly to the period-by-period reward signal and, crucially, provides a continuously updating signal that feeds the termination assessment

defined in Section 4.4.8. A project whose cumulative deviation penalty grows monotonically signals persistent underperformance before the contractual deadline is reached.

**Calibration to contractual terms.** The penalty weight  $\lambda_i$  is calibrated so that the cumulative penalty at the planned finish date, under a sustained schedule delay equal to the contractual delay limit, equals  $\Lambda_i \cdot CP_i$ . This ensures consistency with the delay-penalty exposure that FIDIC-based contracts assign to the contractor, while extending the penalty signal to the full execution window rather than applying it only at completion FIDIC (2017).

The cumulative deviation penalty accrued up to period  $t$  is:

$$\Delta_{i,t}^{\text{pen}} = \sum_{\tau=s_i}^t c_{i,\tau}^{\text{dev}}.$$

This quantity enters both the objective function (as a running cost) and the termination condition (as a trigger threshold).

#### 4.4.8 Termination Conditions and Settlement

A project is terminated at the first period  $t$  at which both of the following conditions hold simultaneously:

1. **Cumulative deviation penalty reaches the cap.** The accrued deviation penalty has reached or exceeded the contractual ceiling:

$$\Delta_{i,t}^{\text{pen}} \geq \kappa_i \cdot \Lambda_i \cdot CP_i,$$

where  $\kappa_i \in (0, 1]$  is the project-specific termination threshold as a fraction of the cap. This condition indicates that the project has exhausted its schedule tolerance and the client has lost contractual leverage over the contractor.

2. **Sustained minimal allocation.** The decision-maker has allocated only the mandatory minimum to the project for  $k$  consecutive periods:

$$x_{i,\tilde{t}} = m_{i,\tilde{t}}^0 \quad \forall \tilde{t} \in \{t - k + 1, \dots, t\},$$

signaling that continued investment is judged economically unattractive.

The simultaneous satisfaction of both conditions ensures that termination occurs only when the project is both contractually disadvantageous (condition 1, driven by the running deviation signal) and operationally abandoned (condition 2, driven by the allocation history). The termination settlement mechanics are defined in Section 4.4.6.

#### 4.4.9 Portfolio Aggregation

The portfolio  $\mathcal{P}$  aggregates  $n$  projects whose individual dynamics are coupled through the shared portfolio cash balance  $B_t$ . The cash balance evolves as:

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t, \quad \forall t \in \mathcal{T},$$

where:

$$V_t = \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{M}_i} R_{i,j}^{\text{net}} \cdot \mathbf{1}\{\text{milestone } (i, j) \text{ newly certified at } t\}.$$

The balance  $B_t \geq 0$  is enforced at all periods; no borrowing or credit facility is available. This hard constraint eliminates the soft budget dynamics present in formulations that allow internal credit and is consistent with the governance structure of a portfolio-level board operating under a fixed capital commitment ?.

The portfolio state at period  $t$  is:

$$S_t = (\{P_{i,t}^{\text{actual}}, E_{i,t}, \text{SPI}_{i,t}, \text{CPI}_{i,t}, \delta_{i,t}, \Delta_{i,t}^{\text{pen}}, \text{status}_{i,t}\}_{i \in \mathcal{P}}, B_t).$$

The dimension of  $S_t$  scales linearly with  $n$ . This linear scaling, combined with the full-horizon coupling through  $B_t$ , characterizes the computational complexity of the portfolio allocation problem and motivates the intractability argument in Level 2.

#### 4.4.10 Objective

The decision-maker seeks to maximize the discounted cumulative net cash flow over the planning horizon:

$$\max_{\{x_{i,t}\}} \mathbb{E} \left[ \sum_{t=1}^H \gamma^{t-1} \left( V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} - \sum_{i \in \mathcal{P}_t^{\text{active}}} c_{i,t}^{\text{dev}} \right) \right].$$

There is no separate terminal sum. Completion incentives are embedded structurally: the retention amount  $R_i^{\text{ret}}$  and the final milestone payment  $R_{i,M_i}^{\text{net}}$  enter through  $V_t$  at the period in which completion is certified, rewarding the decision-maker for bringing projects to completion within the horizon. Projects that are terminated early contribute  $R_i^{\text{term}}$  to  $V_t$  at the termination period, which may be negative, creating a natural deterrent against premature abandonment.

In Level 1, the expectation  $\mathbb{E}[\cdot]$  collapses to a deterministic evaluation under  $\bar{\eta}_{i,t}$ ; the stochastic operator is retained in this shared statement to preserve continuity across all three levels.

The constraints on  $\{x_{i,t}\}$  are:

$$\begin{aligned} x_{i,t} &\geq m_{i,t}^0, & \forall i \in \mathcal{P}_t^{\text{active}}, t \in \mathcal{T}, \\ \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} &\leq B_t, & \forall t \in \mathcal{T}, \\ x_{i,t} &\geq 0, & \forall i \in \mathcal{P}, t \in \mathcal{T}. \end{aligned}$$

The mandatory minimum  $m_{i,t}^0$  reflects the contractual floor below which a project cannot be maintained without triggering a default event; it is calibrated from FIDIC minimum-progress obligations and is now a single fixed parameter per project per period, with no intervention-dependent component.

These primitives — parameters, state variables, cash-flow mechanics, the symmetric deviation penalty, termination conditions, and the shared objective statement — are the foundation from which each of the three formulation levels is constructed. Level 1 specializes them under deterministic full foresight; Level 2 reintroduces stochasticity explicitly; Level 3 reformulates the resulting stochastic program as a CMDP.

## 5 Methodology

### 5.1 Overview of the Proposed Solution Framework

### 5.2 Literature-Calibrated Synthetic Portfolio Generation

### 5.3 Environment Initialization

### 5.4 MDP Specification

### 5.5 State Representation

### 5.6 Action Space and Feasibility Handling

### 5.7 Transition Dynamics

### 5.8 Reward Function

### 5.9 PPO Training Procedure

### 5.10 Benchmark Policies and Evaluation Metrics

### 5.11 Forward Policy Rollout

## 6 Results and Analysis

### 6.1 Overview

*[This section will be completed in future work after experimental implementation. Placeholder structure provided below.]*

### 6.2 Solution Quality Comparison

*[To be completed: Tables and figures showing:]*

- Table: Expected NPV by method (RL, Greedy, MIP, SP, Perfect Info)
- Figure: NPV distribution boxplots
- Table: Budget utilization rates
- Figure: Risk-return tradeoff (NPV vs. variance)

#### **Expected findings:**

- RL outperforms deterministic MIP by  $\sim 15\text{--}20\%$  in expected NPV
- RL matches or exceeds stochastic programming performance
- RL achieves  $\sim 90\text{--}95\%$  of perfect-information upper bound

### 6.3 Computational Performance

*[To be completed: Tables and figures showing:]*

- Table: Solve time by portfolio size and method
- Figure: Scalability plot (log-log scale)

- Table: Inference latency statistics (mean, P95, P99)

**Expected findings:**

- RL inference: 100–200ms per decision (constant with  $N$ )
- MIP solve time: 15–45 minutes for  $N = 10$ , exponential growth
- Speedup factor:  $10,000\times$  for  $N = 10$ , increasing with  $N$

## 6.4 Sensitivity Analysis

*[To be completed: Analysis of performance under varying conditions:]*

**Uncertainty Level:**

- RL maintains robust performance across low/medium/high uncertainty
- Deterministic MIP degrades significantly under high uncertainty
- SP performance depends on scenario tree quality

**Budget Constraint Tightness:**

- RL learns effective prioritization under tight budgets
- Greedy heuristic fails to optimize across projects
- MIP optimal under loose budgets but slow to compute

**Portfolio Composition:**

- Performance consistent across 60/40 domestic/international mix
- Sensitivity to extreme compositions (80/20, 20/80) to be analyzed

## 6.5 Ablation Studies

*[To be completed: Component-wise analysis:]*

- Impact of rolling horizon length  $H \in \{6, 12, 18, 24\}$
- Effect of reward function components (NPV, utilization, penalties)
- Importance of SPI-CPI correlation in training data
- Value of attention mechanism (if implemented)

## 6.6 Discussion

*[To be completed: Interpretation of results, including:]*

**Why RL Outperforms Classical Methods:**

- Learns robust policies from stochastic simulations
- Adapts to partial observability without explicit transition probabilities
- Generalizes across portfolio configurations

**Computational Advantage:**

- One-time training cost amortized over many deployments
- Inference time independent of portfolio size (within trained range)
- Enables real-time scenario analysis and Monte Carlo simulation

**Limitations:**

- Performance bounded by training data distribution
- Requires fine-tuning for company-specific patterns
- Black-box nature may limit practitioner trust (addressed by interpretability analysis)

## 7 Conclusion

### 7.1 Summary of Contributions

This paper establishes the foundational methodology for Reinforcement Learning-based project portfolio budgeting under cashflow uncertainty. Our work addresses a critical gap in operations research: the inability of classical optimization methods to handle sequential decision-making under partial observability in portfolio management.

We make four primary contributions:

**(1) Literature-Calibrated Synthetic Data Generation:** We developed a comprehensive framework for generating realistic project portfolios with empirically-grounded uncertainty distributions. All parameters are calibrated from published studies (318 megaprojects, multiple empirical analyses), ensuring scientific validity while addressing confidentiality constraints.

**(2) Rolling Horizon Control Architecture:** We proposed a rolling horizon framework that maintains fixed 12-period episodes while handling variable-duration portfolios through sequential re-planning, preventing state-space explosion and enabling tractable RL training.

**(3) Formal Problem Formulation:** We provided the first rigorous mathematical formulation of portfolio budgeting as a POMDP with explicit state space, action space, transition dynamics, and reward function, all grounded in empirical project management literature.

**(4) Complete Modeling Framework:** We documented a modular portfolio modeling system covering composition, cashflows, uncertainty, and revenue plans, with all assumptions explicitly stated and future extensions identified.

### 7.2 Implications for Research and Practice

**For Operations Research:** This work demonstrates that RL can address problems where classical OR methods fail due to unknown transition probabilities and computational intractability. The rolling horizon architecture provides a template for applying RL to other long-horizon sequential decision problems.

**For Project Management:** The literature-calibrated synthetic data methodology enables reproducible research in domains with confidentiality constraints. Our framework can be extended to construction, infrastructure, R&D, and other project-intensive industries.

**For Machine Learning:** This application showcases RL’s potential in operations research domains, complementing recent successes in robotics, game playing, and control. The integration of domain knowledge (S-curves, milestone payments, SPI/CPI dynamics) with data-driven learning exemplifies hybrid AI approaches.

### 7.3 Limitations and Scope Boundaries

We deliberately adopted the *Simplest Model Principle* to establish a foundational framework. Key limitations include:

- **Aggregated cashflow modeling:** No resource-level detail (labor, materials, equipment)
- **Fixed portfolio composition:** 60/40 domestic/international mix assumed optimal
- **Milestone-based payments only:** Alternative structures (time-based, cost-plus) excluded
- **No project interdependencies:** Shared resources and technology spillovers not modeled
- **Synthetic data only:** Real-world validation pending company partnerships

Each limitation represents a natural extension for future work, documented with clear research directions.

### 7.4 Future Work

#### Immediate Next Steps (Current Research):

1. **RL Agent Implementation:** Select and train policy network (PPO, SAC, or Transformer-based)
2. **Baseline Comparison:** Implement MIP, stochastic programming, and greedy heuristics
3. **Experimental Evaluation:** Conduct comprehensive experiments across scenarios
4. **Sensitivity Analysis:** Analyze robustness to parameter variations

#### Medium-Term Extensions (6–12 Months):

1. **Fine-Tuning Protocol:** Develop methodology for adapting pretrained models to company-specific data
2. **Interpretability Analysis:** Apply attention mechanisms or SHAP values to explain allocation decisions
3. **Multi-Objective Optimization:** Extend to Pareto-optimal policies balancing NPV, risk, and strategic alignment
4. **Real-World Validation:** Partner with EPC contractors for pilot deployment

#### Long-Term Research Directions (1–3 Years):

1. **Resource-Level Modeling:** Disaggregate cashflows into labor, materials, equipment with detailed scheduling
2. **Dynamic Portfolio Selection:** Integrate project selection with execution optimization
3. **Multi-Agent RL:** Model interactions between contractor, client, and subcontractors
4. **Transfer Learning:** Investigate cross-domain transfer (oil & gas  $\rightarrow$  construction  $\rightarrow$  infrastructure)
5. **Hierarchical RL:** Develop two-level policies (strategic portfolio allocation + tactical project control)

## 7.5 Broader Impact

This research has potential impact beyond project portfolio management:

### Methodological Contributions:

- Template for applying RL to OR problems with unknown transition dynamics
- Framework for literature-calibrated synthetic data generation
- Rolling horizon architecture for long-horizon sequential decision problems

### Practical Applications:

- Real-time decision support for portfolio managers
- Scenario analysis for strategic planning
- Risk assessment through Monte Carlo simulation
- Training tool for project management education

### Open Science:

- Complete codebase and datasets released as open source
- Reproducible research enabling community validation and extension
- Benchmark problem for RL algorithm development

## 7.6 Concluding Remarks

Project portfolio budgeting under uncertainty represents a critical challenge for capital-intensive industries. Traditional optimization methods fail due to computational intractability and unknown state transition probabilities. This paper establishes the foundational methodology for addressing this challenge through Reinforcement Learning with rolling horizon control.

By developing a literature-calibrated synthetic data generation framework, we overcome the confidentiality barrier that has prevented progress in this domain. Our formal problem formulation and complete modeling framework provide a rigorous foundation for future research.

The path forward is clear: implement and evaluate RL agents, compare against classical baselines, and develop fine-tuning protocols for company-specific deployment. We invite the research community to build upon this foundation, extending the methodology to related domains and advancing the state of the art in RL-based operations research.

The complete modeling framework, instance generator, and documentation will be released as open-source software upon publication, enabling reproducible research and practical deployment in this emerging field.

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## A Mathematical Proofs

*[Placeholder for mathematical proofs. To be completed if needed.]*

### A.1 Proof of Rolling Horizon Optimality Conditions

*[To be added: Conditions under which rolling horizon policy approximates infinite-horizon optimal policy]*

### A.2 Proof of State Space Completeness

*[To be added: Demonstration that state representation captures all decision-relevant information]*

## B Parameter Calibration Details

### B.1 Complete Parameter Tables

This appendix provides comprehensive parameter tables for all model components, with literature citations and calibration details.

#### B.1.1 Portfolio Composition Parameters

Table 2: Portfolio Composition Model Parameters

| Parameter            | Value | Unit     | Source                    |
|----------------------|-------|----------|---------------------------|
| Domestic weight      | 0.60  | fraction | Khanzadi et al. (2018)    |
| International weight | 0.40  | fraction | Khanzadi et al. (2018)    |
| Portfolio size min   | 8     | projects | Merrow (2011), CII (2019) |
| Portfolio size max   | 18    | projects | Merrow (2011), CII (2019) |

#### B.1.2 Profit Margin Parameters

Table 3: Profit Margin Distribution Parameters

| Parameter        | Domestic | International | Source                 |
|------------------|----------|---------------|------------------------|
| Mean $\mu$       | 0.10     | 0.14          | Khanzadi et al. (2018) |
| Std Dev $\sigma$ | 0.012    | 0.030         | Khanzadi et al. (2018) |
| Lower bound $a$  | 0.08     | 0.12          | Iran Budget Law        |
| Upper bound $b$  | 0.12     | 0.16          | Market analysis        |

Table 4: Project BAC Lognormal Distribution

| Parameter       | Value  | Source            |
|-----------------|--------|-------------------|
| $\mu_{\log}$    | 5.01   | Merrow (2011)     |
| $\sigma_{\log}$ | 0.69   | Merrow (2011)     |
| Median BAC      | \$150M | Merrow (2011)     |
| Mean BAC        | \$280M | Merrow (2011)     |
| Min BAC         | \$50M  | Industry practice |
| Max BAC         | \$5B   | Industry practice |

Table 5: S-Curve Beta Distribution Base Parameters

| Parameter               | Value | Source                 |
|-------------------------|-------|------------------------|
| $\alpha_{\text{base}}$  | 2.5   | Cioffi (2005)          |
| $\beta_{\text{base}}$   | 2.0   | Cioffi (2005)          |
| Peak time $\tau^*$      | 0.60  | Kenley & Wilson (1986) |
| Size exponent           | 0.15  | Barraza et al. (2007)  |
| Duration exponent       | 0.12  | Barraza et al. (2007)  |
| Residual noise $\sigma$ | 0.10  | Calibrated             |

### B.1.3 Project BAC Parameters

### B.1.4 S-Curve Parameters

### B.1.5 Performance Uncertainty Parameters

### B.1.6 Duration Model Parameters

### B.1.7 Milestone Payment Structure

## B.2 Validation Results

*[To be completed: Statistical validation of generated instances against target distributions]*

### B.2.1 Distributional Validation

*[Tables showing sample statistics vs. target values for all key variables]*

### B.2.2 Correlation Validation

*[Correlation matrices comparing generated data to empirical values]*

### B.2.3 Portfolio-Level Validation

*[Aggregate statistics for generated portfolios]*

## C Instance Generator Algorithm

### C.1 Instance Generator Algorithm

This appendix provides the complete pseudocode for the portfolio instance generator.

Table 6: SPI and CPI Distribution Parameters

| Parameter               | SPI  | CPI  | Correlation | Source             |
|-------------------------|------|------|-------------|--------------------|
| $\alpha$                | 2.8  | 3.2  | –           | Morrow (2011)      |
| $\beta$                 | 2.2  | 1.8  | –           | Love et al. (2016) |
| Lower bound             | 0.3  | 0.5  | –           | Empirical range    |
| Upper bound             | 1.3  | 1.2  | –           | Empirical range    |
| Mean                    | 0.86 | 0.87 | –           | Calibrated         |
| Std Dev                 | 0.18 | 0.14 | –           | Calibrated         |
| $\rho(\text{SPI, CPI})$ | –    | –    | 0.68        | Love et al. (2016) |

Table 7: Project Duration Model Parameters

| Parameter                 | Value     | Source            |
|---------------------------|-----------|-------------------|
| $D_{\text{base}}$         | 24 months | Morrow (2011)     |
| $\text{BAC}_{\text{ref}}$ | \$150M    | Morrow (2011)     |
| Scale exponent            | 0.35      | Morrow (2011)     |
| Noise $\sigma_D$          | 0.15      | Calibrated        |
| Min duration              | 12 months | Industry practice |
| Max duration              | 60 months | Industry practice |

**Algorithm 1** Portfolio Instance Generator**Require:** Portfolio size range  $[N_{\min}, N_{\max}]$ , parameter tables**Ensure:** Portfolio instance  $\mathcal{P}$  with all attributes

```

1:
2: Stage 1: Portfolio Composition
3:  $N \leftarrow \text{UniformInt}(N_{\min}, N_{\max})$ 
4:  $N_{\text{domestic}} \leftarrow \lfloor 0.6 \times N \rfloor$ 
5:  $N_{\text{international}} \leftarrow N - N_{\text{domestic}}$ 
6:
7: Stage 2: Project Characteristics
8: for  $i = 1$  to  $N$  do
9:   Assign category: domestic or international
10:  Sample  $\log(\text{BAC}_i) \sim \mathcal{N}(\mu_{\log}, \sigma_{\log})$ 
11:  Clip  $\text{BAC}_i$  to  $[\$50M, \$5B]$ 
12:  Sample  $\pi_i \sim \text{TruncatedNormal}(\mu_{\pi}, \sigma_{\pi}, a_{\pi}, b_{\pi})$ 
13:  Compute  $R_i \leftarrow \text{BAC}_i \times (1 + \pi_i)$ 
14:  Sample  $D_i \leftarrow D_{\text{base}} \times (\text{BAC}_i / \text{BAC}_{\text{ref}})^{0.35} \times \epsilon_D$ 
15:  Clip  $D_i$  to  $[12, 60]$  months
16: end for
17:
18: Stage 3: S-Curve Parameters
19: for  $i = 1$  to  $N$  do
20:   Compute size multiplier:  $m_{\text{size}} \leftarrow (\text{BAC}_i / \text{BAC}_{\text{median}})^{0.15}$ 
21:   Compute duration multiplier:  $m_{\text{dur}} \leftarrow (D_i / D_{\text{median}})^{0.12}$ 
22:   Get category multiplier  $m_{\text{cat}}$  (0.9 for domestic, 1.1 for international)
23:   Sample risk level: low or high
24:   Get risk multipliers  $m_{\text{risk}}^{\alpha}, m_{\text{risk}}^{\beta}$ 
25:   Sample  $\epsilon_{\alpha}, \epsilon_{\beta} \sim \text{LogNormal}(0, 0.10)$ 
26:    $\alpha_i \leftarrow \alpha_{\text{base}} \times m_{\text{size}} \times m_{\text{cat}} \times m_{\text{risk}}^{\alpha} \times \epsilon_{\alpha}$ 
27:    $\beta_i \leftarrow \beta_{\text{base}} \times m_{\text{dur}} \times m_{\text{cat}} \times m_{\text{risk}}^{\beta} \times \epsilon_{\beta}$ 
28:   Clip  $\alpha_i$  to  $[1.2, 5.0]$ ,  $\beta_i$  to  $[1.2, 5.5]$ 
29: end for
30:
31: Stage 4: Cashflow Profiles
32: for  $i = 1$  to  $N$  do

```

Table 8: Standard Milestone Payment Weights

| Milestone   | Progress Threshold | Payment Weight |
|-------------|--------------------|----------------|
| Advance     | 0%                 | 10%            |
| Milestone 1 | 25%                | 20%            |
| Milestone 2 | 50%                | 30%            |
| Milestone 3 | 75%                | 30%            |
| Final       | 100%               | 10%            |

## C.2 Validation Algorithm

---

### Algorithm 2 Instance Validation

---

**Require:** Generated portfolio instance  $\mathcal{P}$ , target statistics

**Ensure:** Validation report (pass/fail for each check)

```

1:
2: Level 1: Distributional Validation
3: for each variable  $X$  in  $\{\text{BAC}, \pi, D, \alpha, \beta, \text{SPI}, \text{CPI}\}$  do
4:   Compute sample mean  $\bar{X}$ , std dev  $s_X$ 
5:   Check:  $|\bar{X} - \mu_{\text{target}}|/\mu_{\text{target}} < 0.05$  ▷ 5% tolerance
6:   Check:  $|s_X - \sigma_{\text{target}}|/\sigma_{\text{target}} < 0.10$  ▷ 10% tolerance
7: end for
8:
9: Level 2: Correlation Validation
10: Compute  $\rho(\text{SPI}, \text{CPI})$ 
11: Check:  $|\rho - 0.68| < 0.05$ 
12:
13: Level 3: Portfolio-Level Validation
14: Compute total portfolio BAC:  $\sum_{i=1}^N \text{BAC}_i$ 
15: Compute domestic fraction:  $\sum_{i \in \text{domestic}} \text{BAC}_i / \sum_{i=1}^N \text{BAC}_i$ 
16: Check: domestic fraction  $\in [0.55, 0.65]$  ▷ 60%  $\pm$  5%
17:
18: Level 4: Sanity Checks
19: for  $i = 1$  to  $N$  do
20:   Check:  $\text{BAC}_i > 0, D_i > 0, \pi_i > 0$ 
21:   Check:  $\alpha_i \in [1.2, 5.0], \beta_i \in [1.2, 5.5]$ 
22:   Check:  $\text{SPI}_i \in [0.3, 1.3], \text{CPI}_i \in [0.5, 1.2]$ 
23:   Check:  $\sum_m w_m = 1.0$  ▷ Milestone weights sum to 100%
24: end for
25:
26: return Validation report

```

---

## C.3 Implementation Notes

**Computational Complexity:** The instance generator has time complexity  $O(N \times D_{\max})$  where  $N$  is portfolio size and  $D_{\max}$  is maximum project duration. For typical values ( $N = 15, D_{\max} = 60$ ), generation takes  $< 1$  second per instance.

**Reproducibility:** All random number generation uses seeded PRNGs (NumPy default\_rng with explicit seeds). Seeds are logged for each generated instance.

**Parallelization:** Instance generation is embarrassingly parallel. For large datasets (10,000+ instances), use multiprocessing with one worker per CPU core.

**Output Format:** Instances are serialized as JSON or HDF5 files with standardized schema for interoperability.